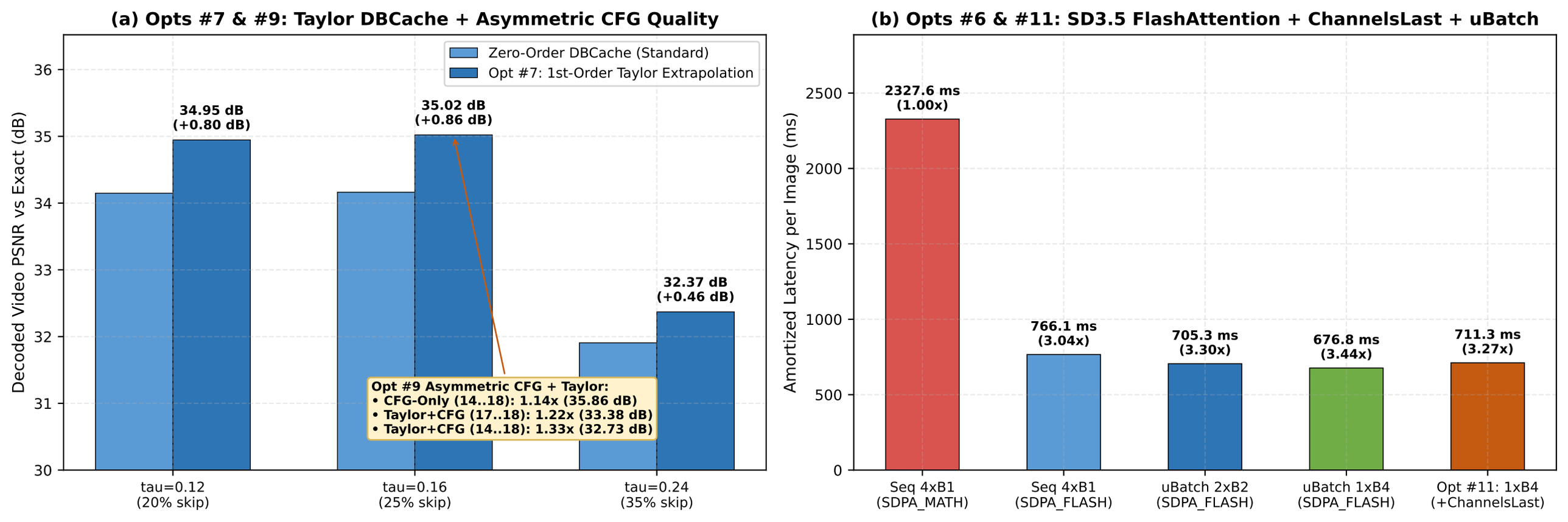
(a) Opts #7 & #9: Taylor DBCache + Asymmetric CFG Quality
Zero-Order DBCache (Standard)
Opt #7: 1st-Order Taylor Extrapolation
Decoded Video PSNR vs Exact (dB)
36
35
34
33
32
31
30
34.95 dB
(+0.80 dB)
35.02 dB
(+0.86 dB)
32.37 dB
(+0.46 dB)
Opt #9 Asymmetric CFG + Taylor:
• CFG-Only (14..18): 1.14x (35.86 dB)
• Taylor+CFG (17..18): 1.22x (33.38 dB)
• Taylor+CFG (14..18): 1.33x (32.73 dB)
tau=0.12
(20% skip)
tau=0.16
(25% skip)
tau=0.24
(35% skip)
(b) Opts #6 & #11: SD3.5 FlashAttention + ChannelsLast + uBatch
Amortized Latency per Image (ms)
2500
2000
1500
1000
500
0
2327.6 ms
(1.00x)
766.1 ms
(3.04x)
705.3 ms
(3.30x)
676.8 ms
(3.44x)
711.3 ms
(3.27x)
Seq 4xB1
(SDPA_MATH)
Seq 4xB1
(SDPA_FLASH)
uBatch 2xB2
(SDPA_FLASH)
uBatch 1xB4
(SDPA_FLASH)
Opt #11: 1xB4
(+ChannelsLast)